

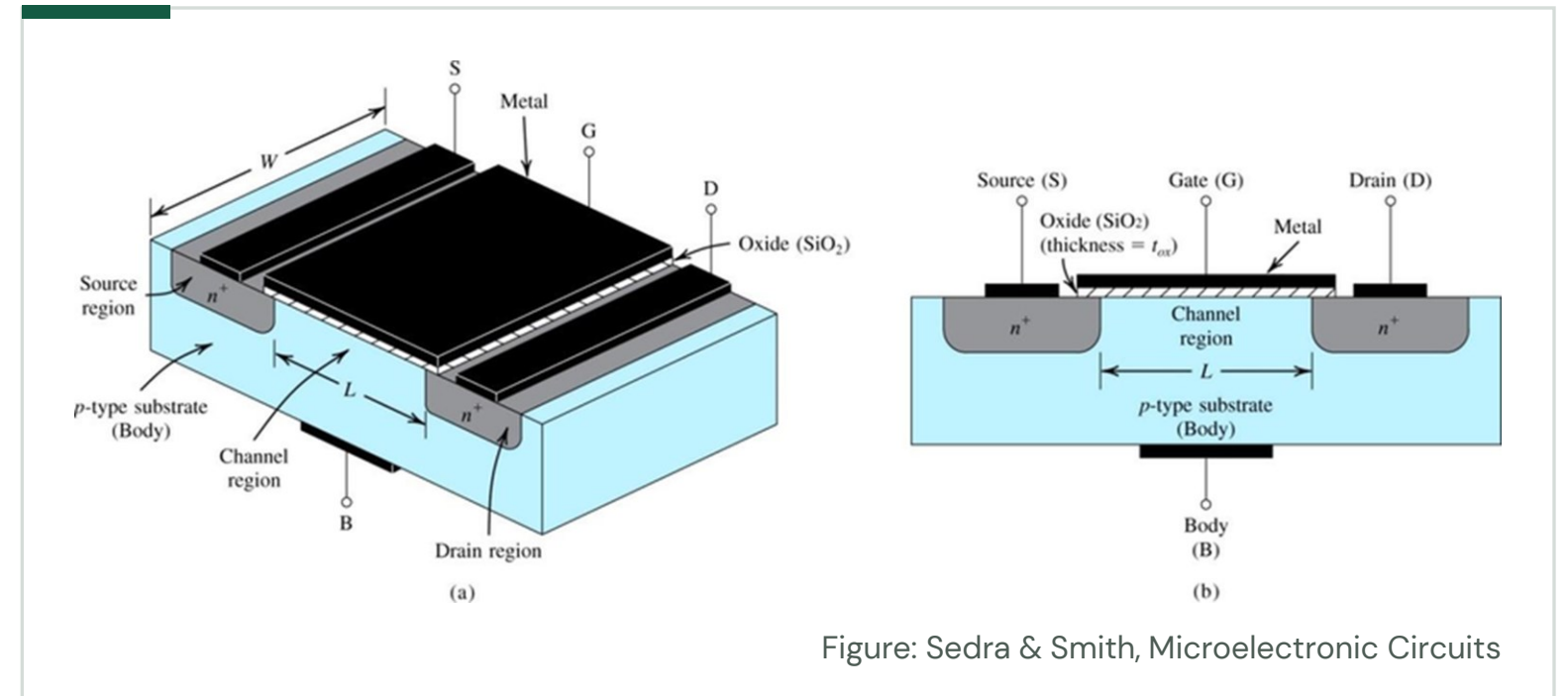
MOSFETS

Metal-oxide-semiconductor
field-effect transistors

LECTURE 3

Dr. Zahraa Ismail

Computer Engineering · Academic Year 2025/2026 · Semester One



LECTURE MAP

1

Structure and operation

how a channel forms; the threshold voltage

2

I–V characteristics

cutoff, triode and saturation

3

MOSFETs in DC circuits

finding the operating point

4

The MOSFET as an amplifier

transconductance, small-signal model

5

The MOSFET as a switch

BY THE END YOU CAN

- Explain how v_{GS} creates a channel and how v_{DS} pinches it off
- Pick the right region and use its i_D equation
- Find the DC operating point of a MOSFET circuit

02

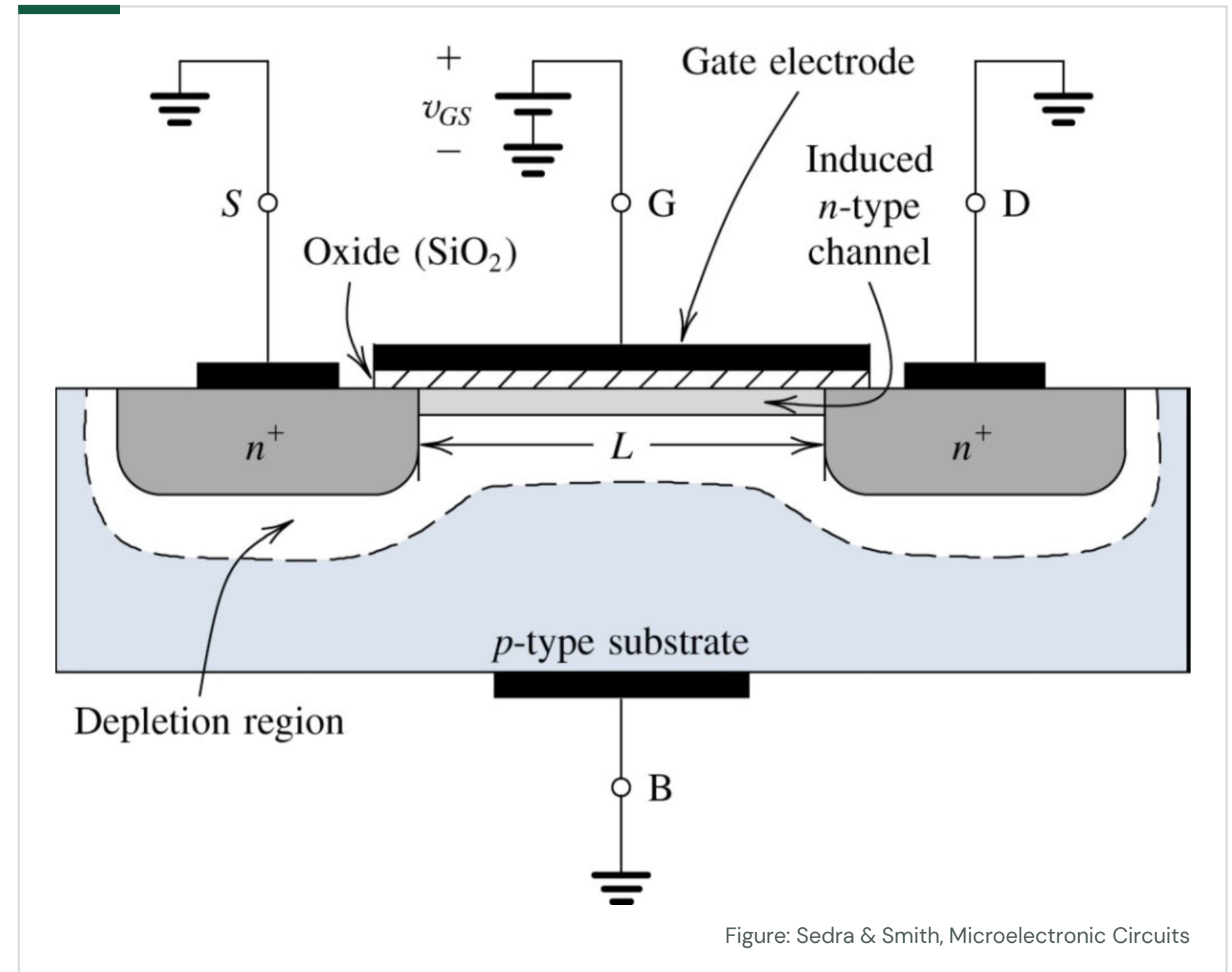
I-V CHARACTERISTICS

- Increasing v_{DS} : pinch-off
- Triode and saturation
- The i_D - v_{DS} family of curves

- 1 Structure and operation
- 2 I-V characteristics**
- 3 MOSFETs in DC circuits
- 4 The MOSFET as an amplifier
- 5 The MOSFET as a switch

CREATING A CHANNEL: THE THRESHOLD VOLTAGE

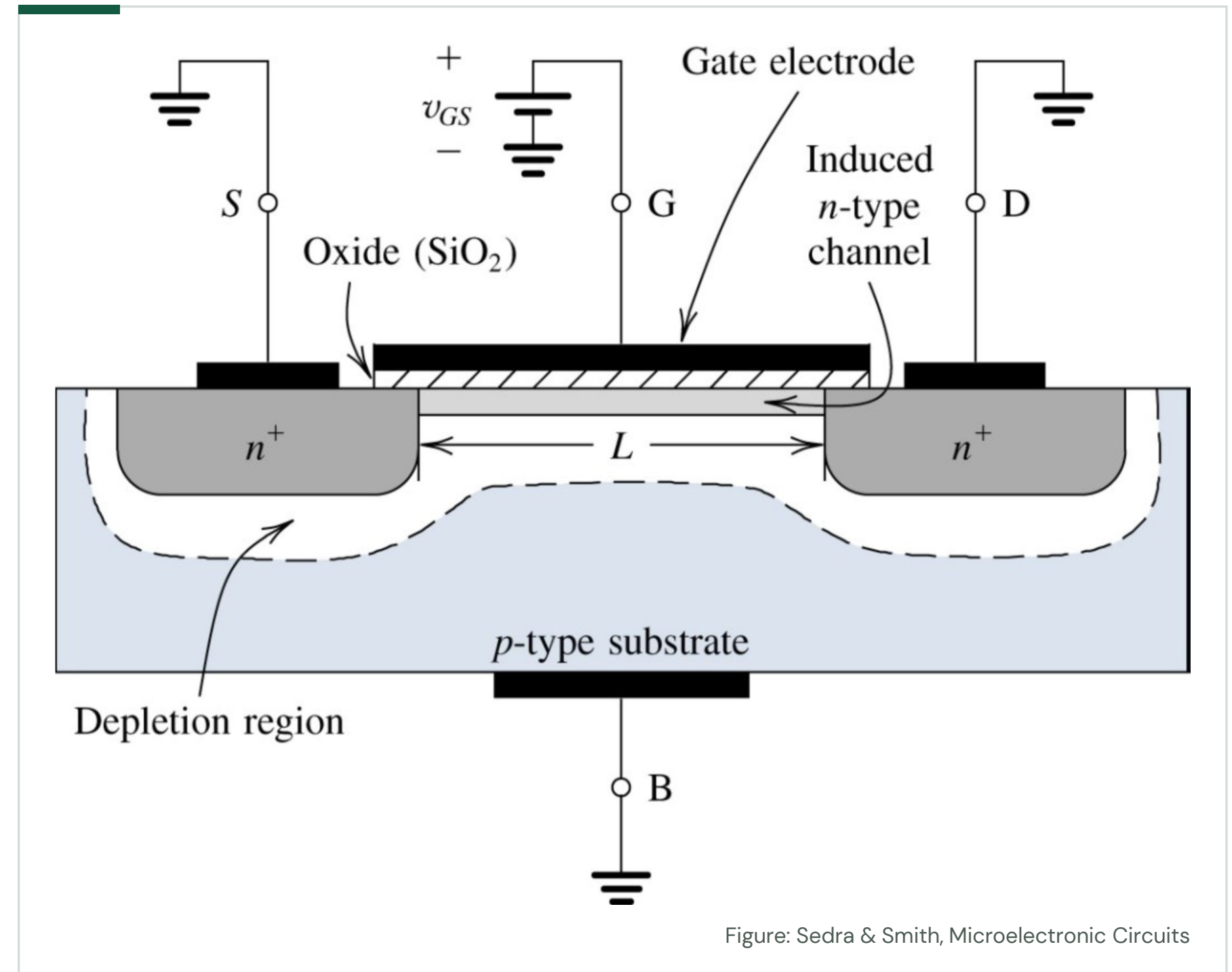
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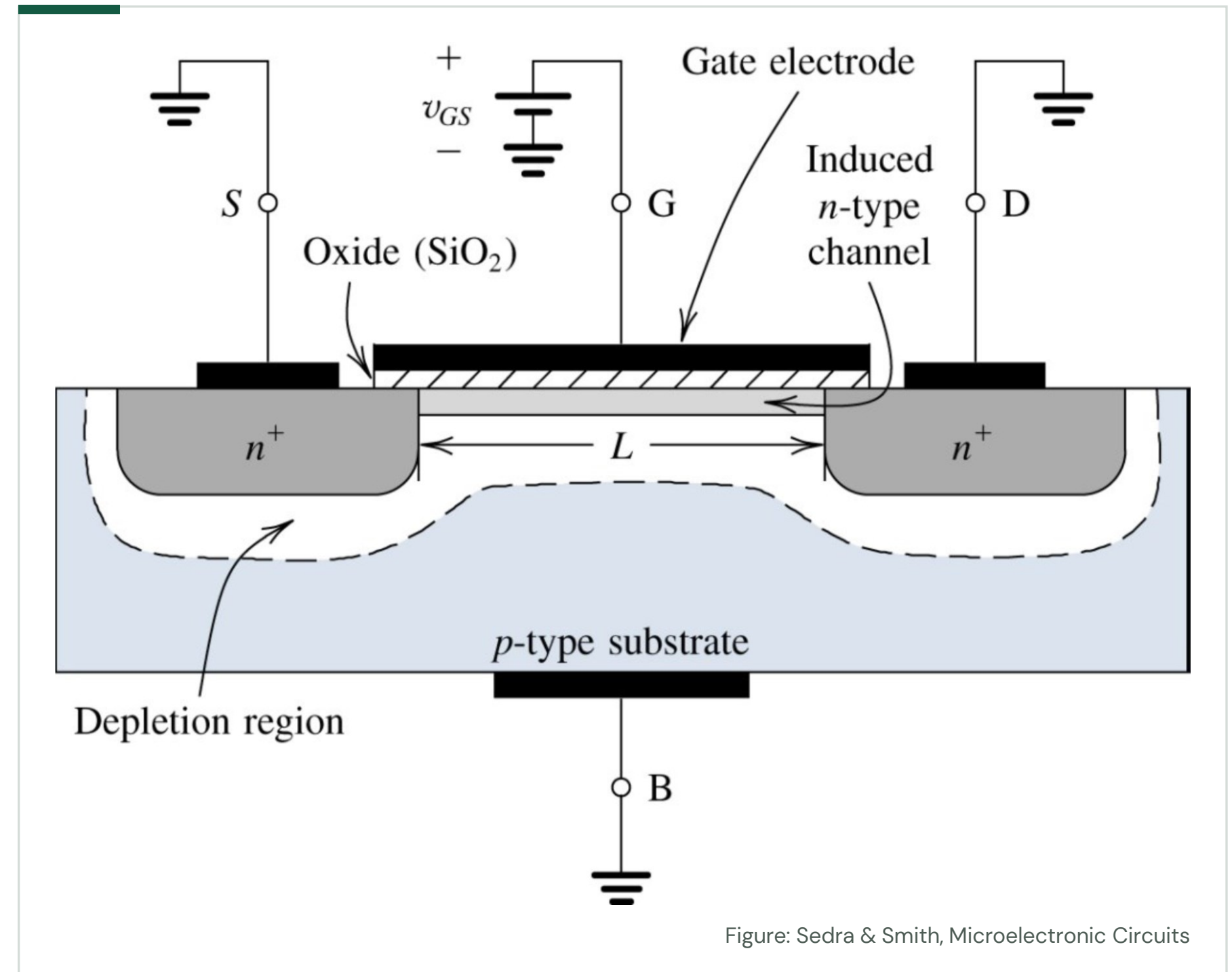
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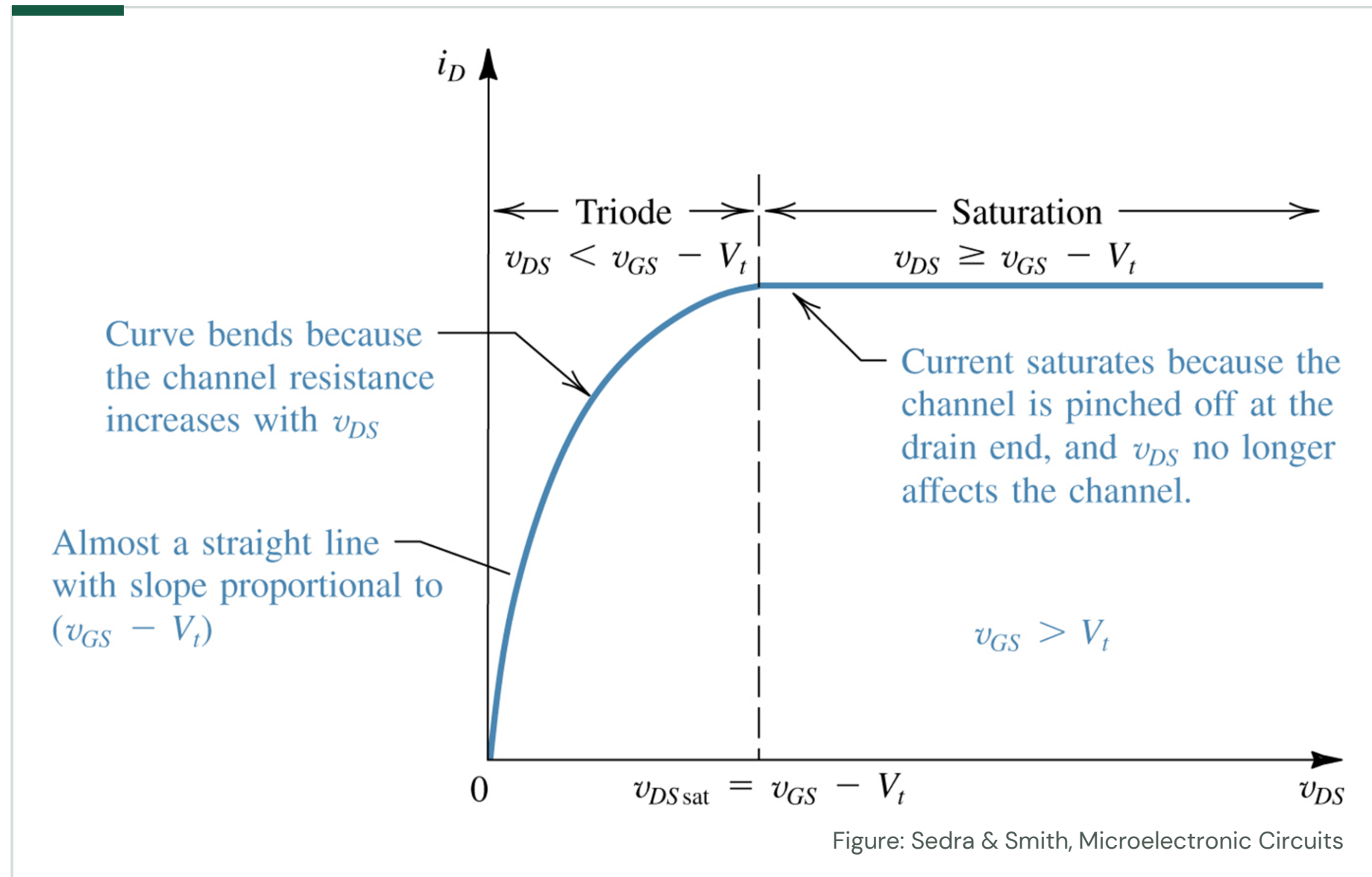
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DEFINITION

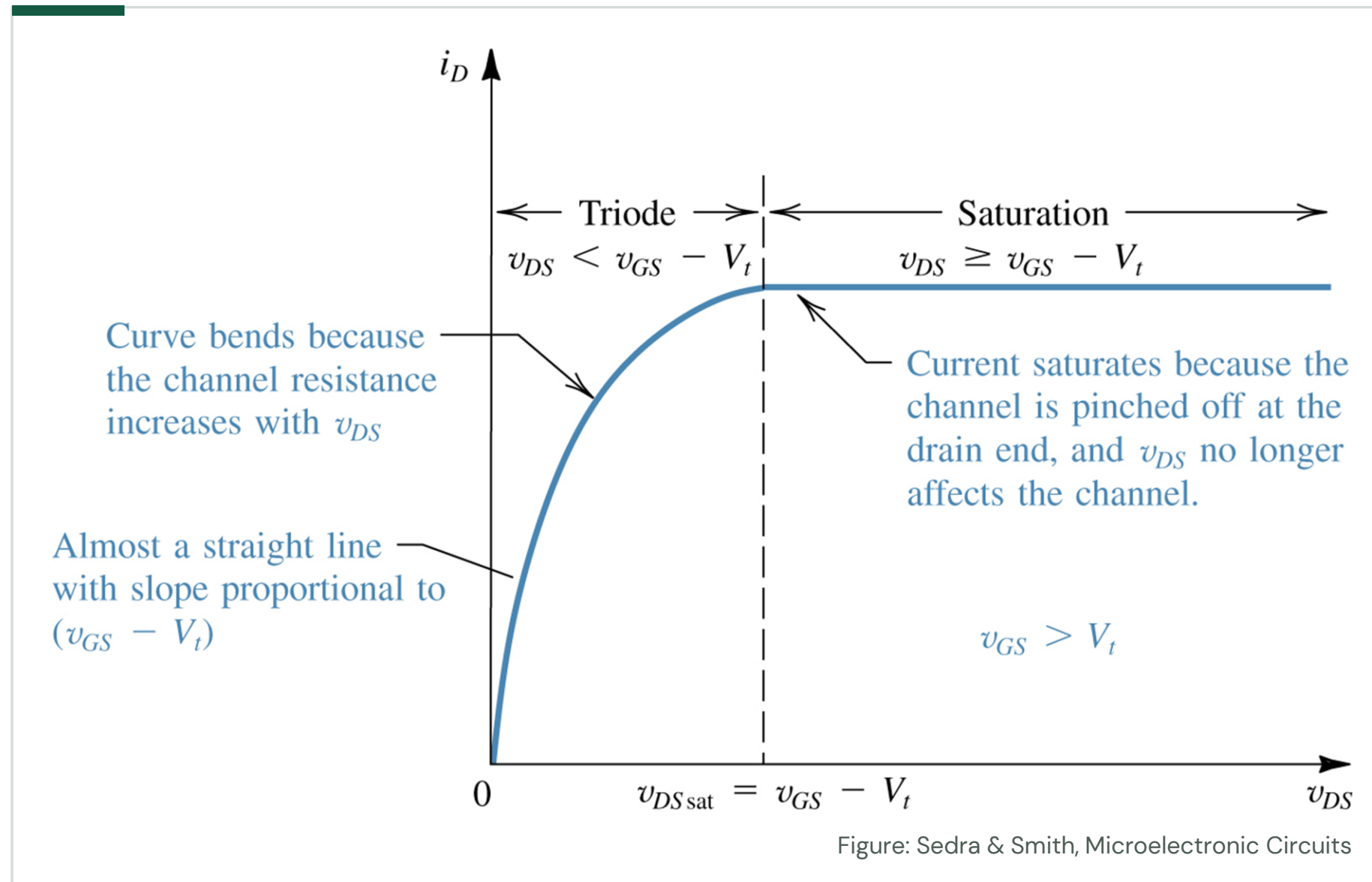
Threshold voltage V_t : the value of v_{GS} at which the channel just forms.



TRIODE AND SATURATION: ONE CURVE, TWO EQUATIONS



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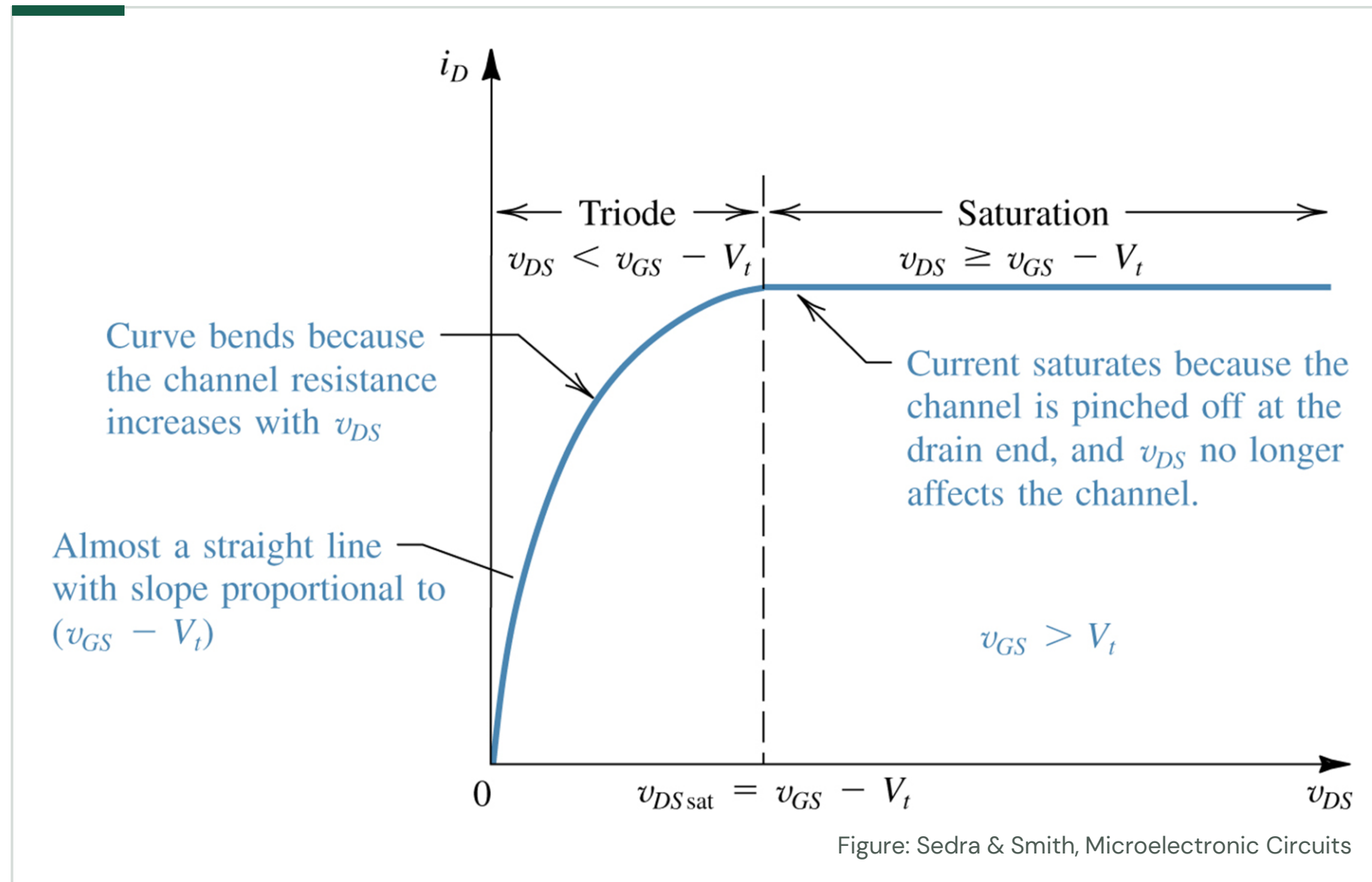


TRIODE

$$v_{DS} < v_{GS} - V_t$$

$$i_D = k'_n \frac{W}{L} \left[(v_{GS} - V_t) v_{DS} - \frac{1}{2} v_{DS}^2 \right]$$

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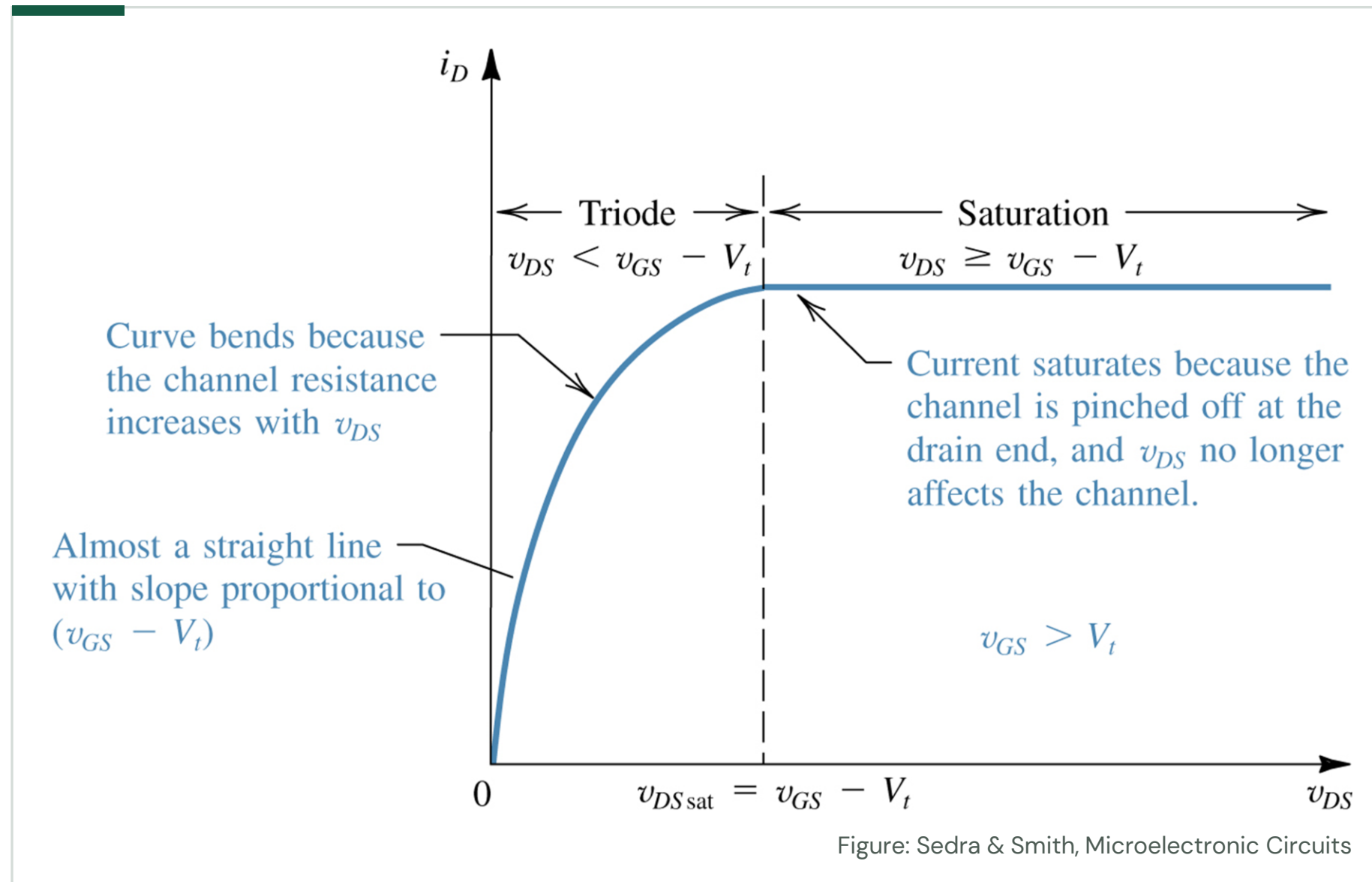
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SATURATION

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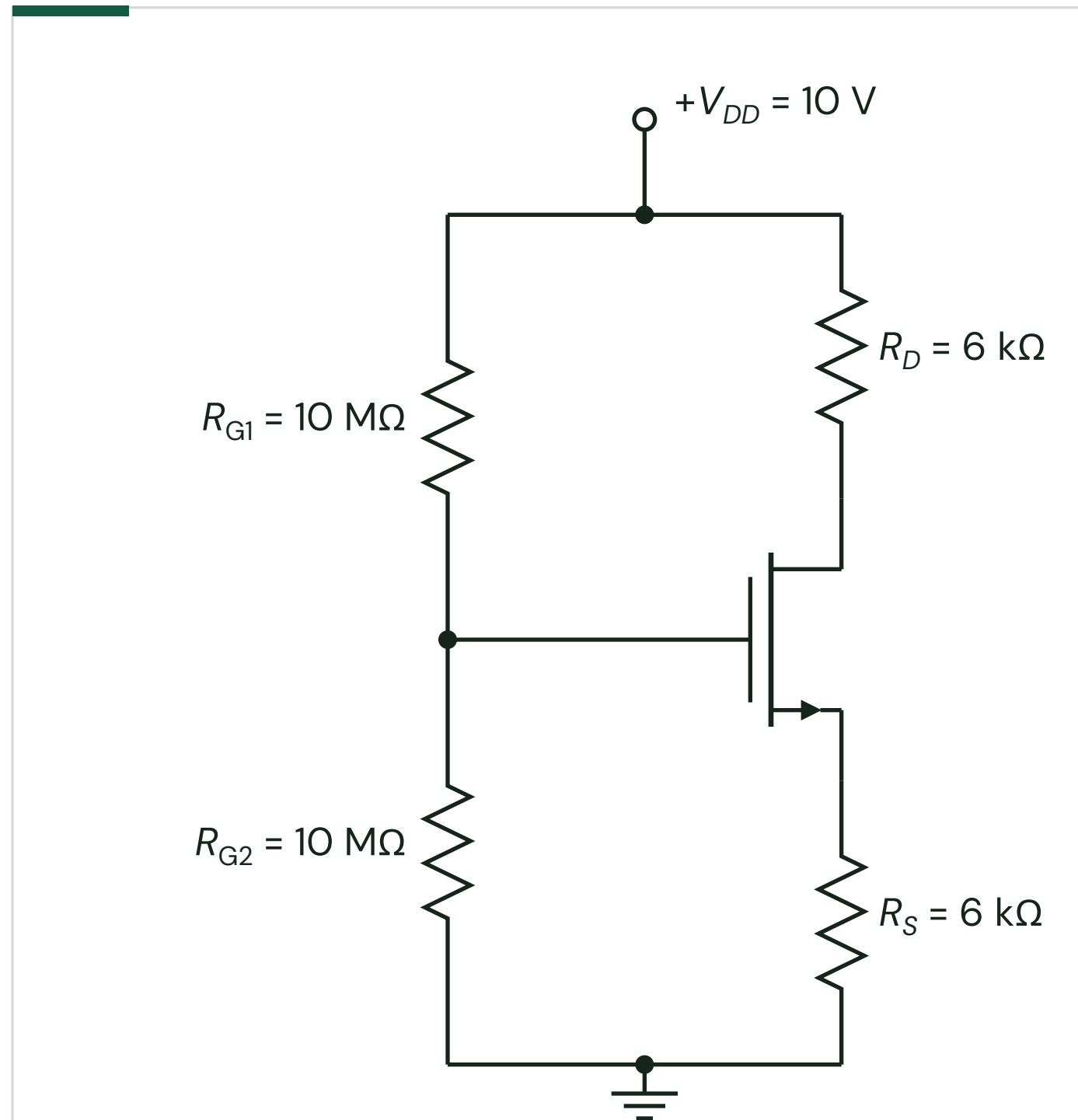
SATURATION $v_{DS} \geq v_{GS} - V_t$

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Both require $v_{GS} > V_t$; without a channel, $i_D = 0$ (cutoff).

EXAMPLE: FIND THE DRAIN VOLTAGE

Given $V_t = 1 \text{ V}$ and $k'_n(W/L) = 1 \text{ mA/V}^2$. The gate draws no current.

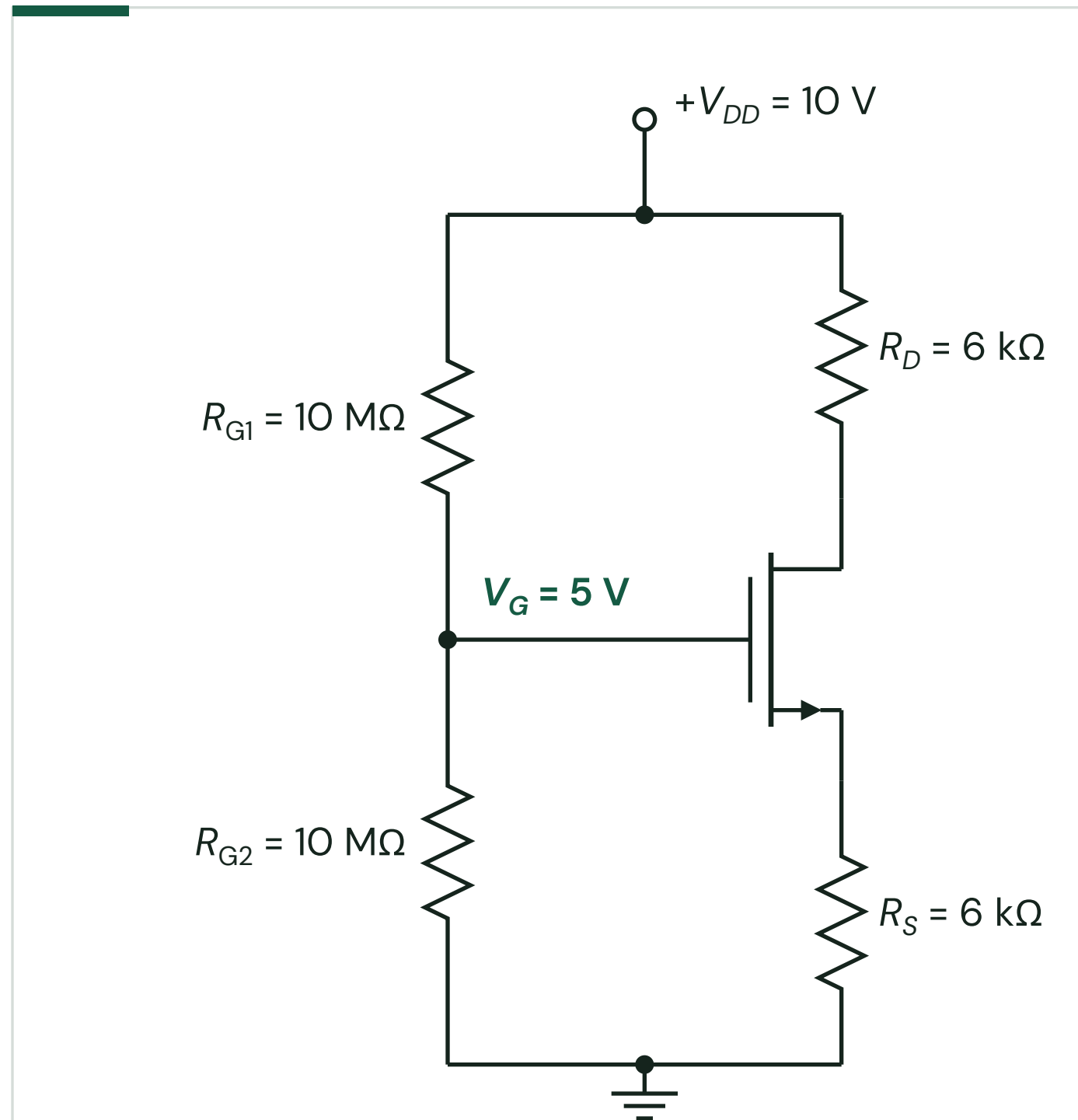


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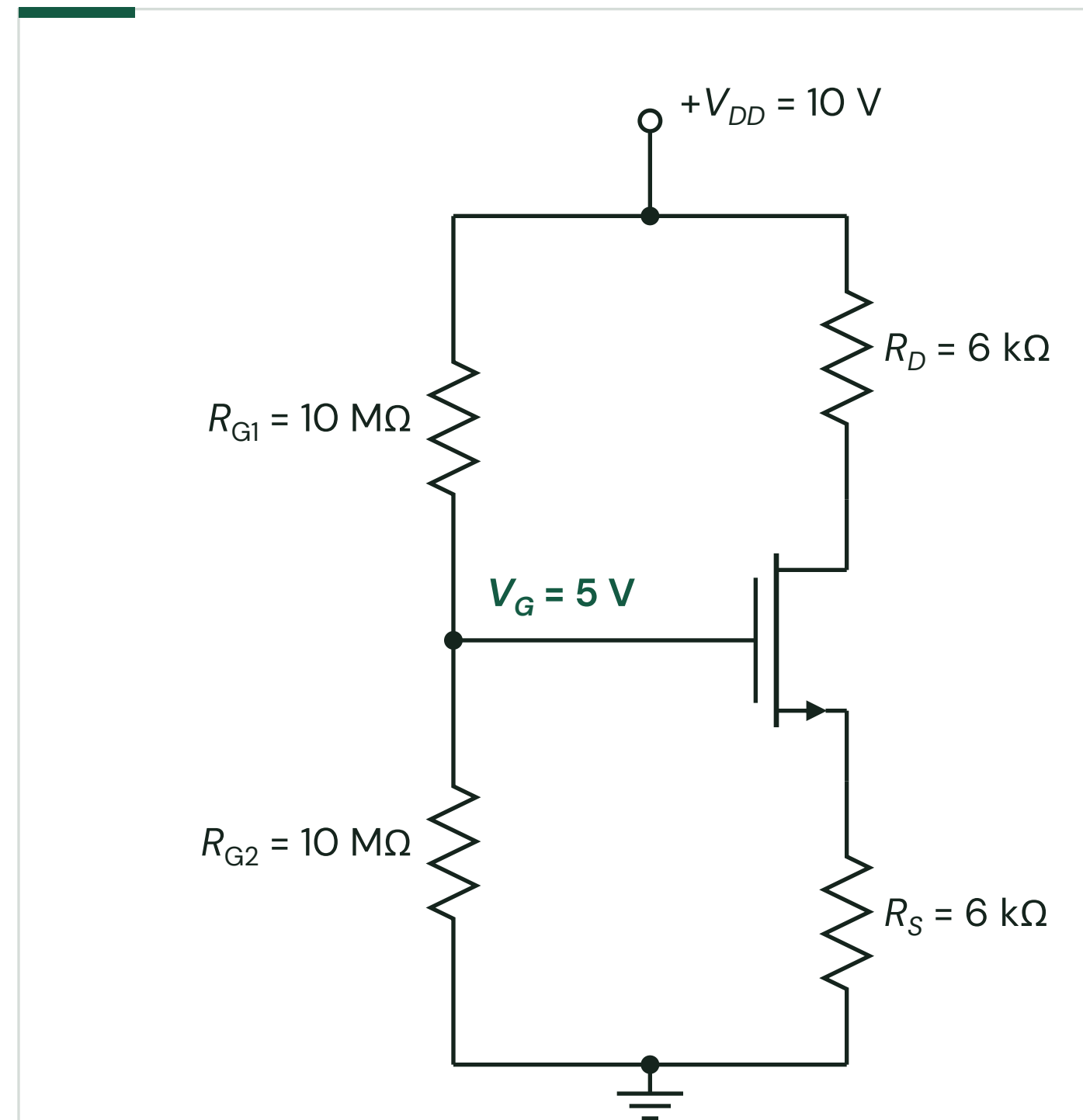
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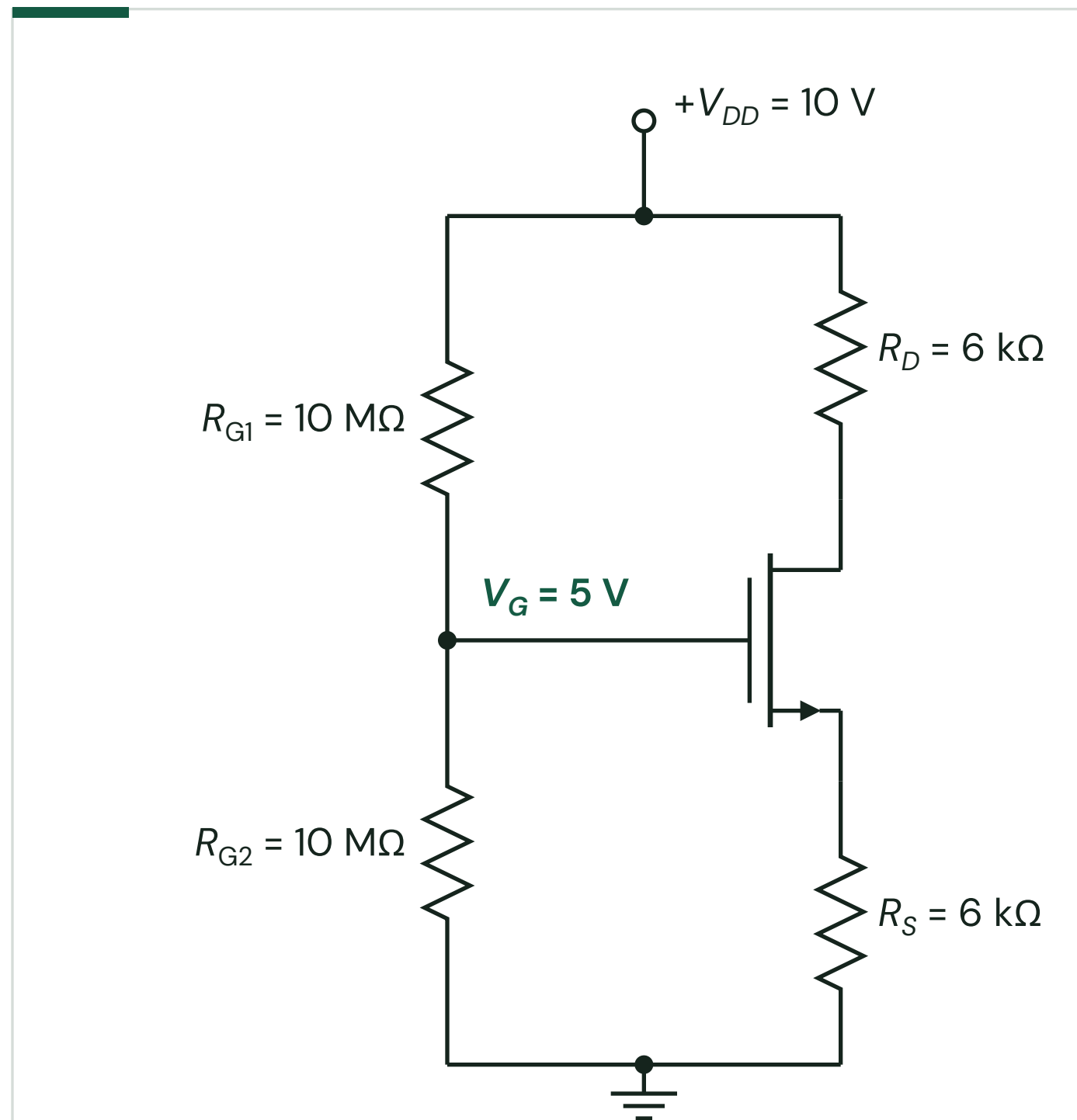
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2 Assume saturation, with $V_{GS} = V_G - I_D R_S$ (I in mA, R in $\text{k}\Omega$)

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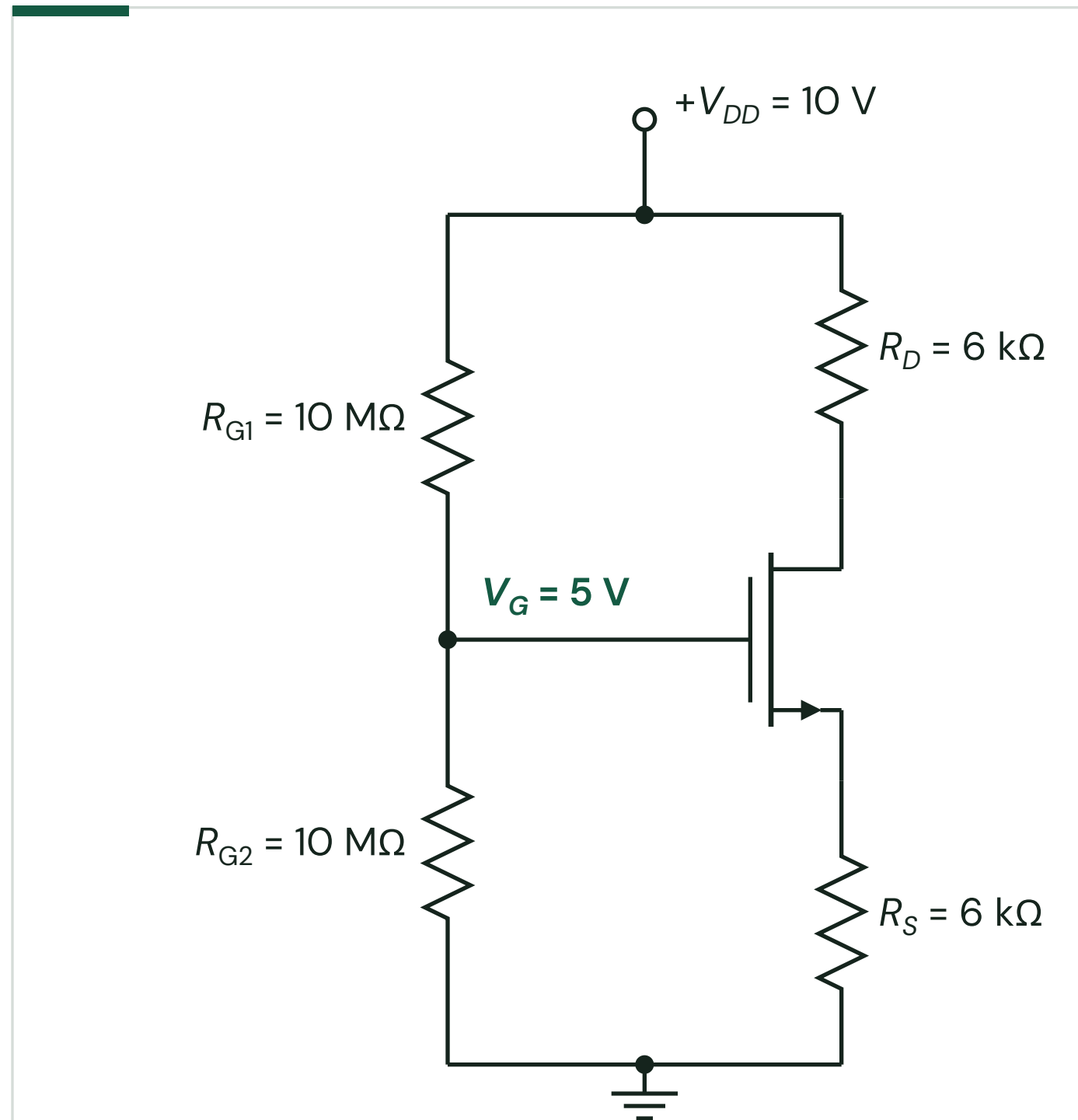
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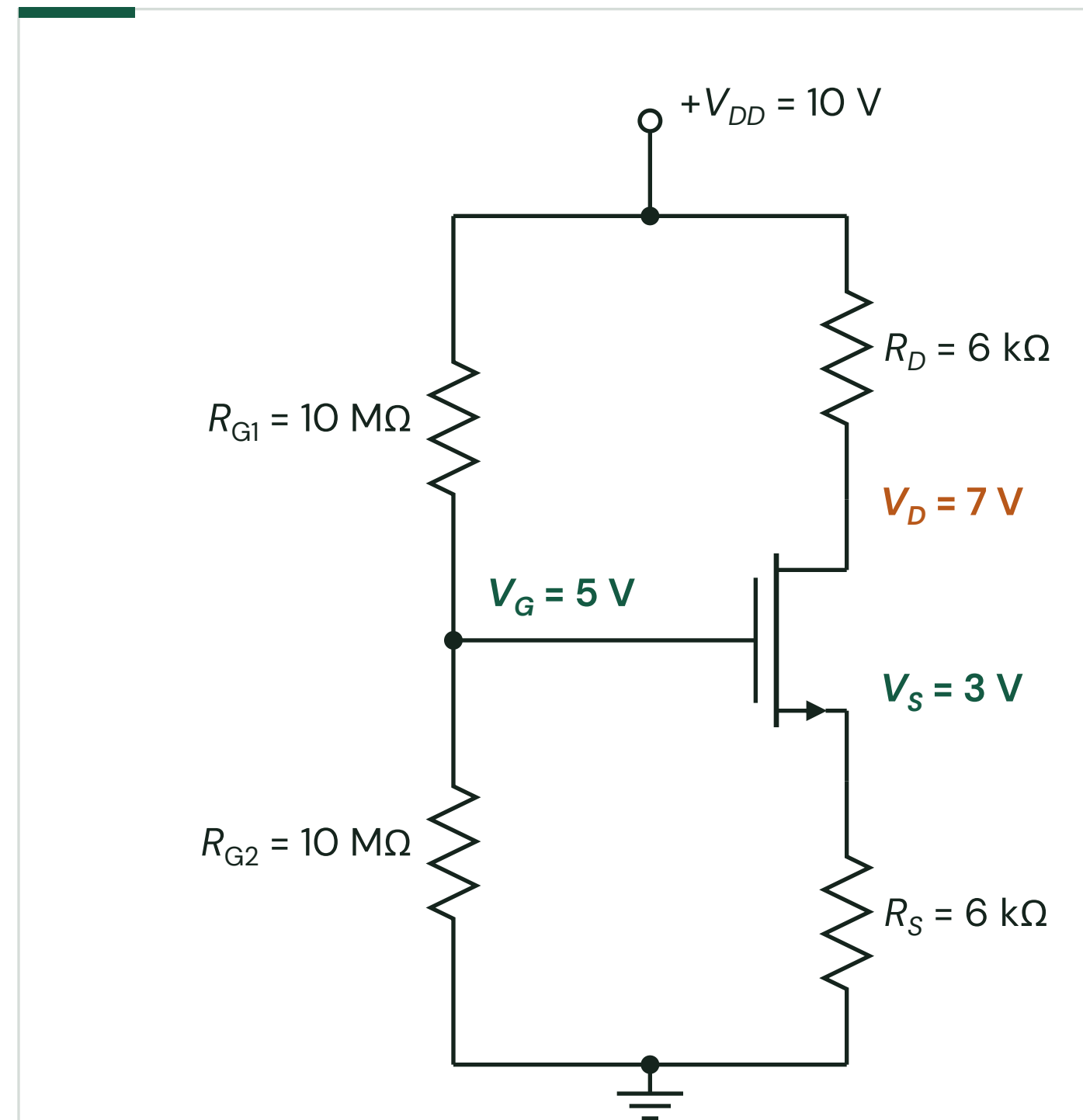
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- 5 $V_D = V_{DD} - I_D R_D = 10 - 0.5(6) = 7 \text{ V}$

$$V_D = 7 \text{ V}$$

✓ $V_{DS} = 4 \text{ V} \geq V_{GS} - V_t = 1 \text{ V}$: saturation holds.

SUMMARY: THREE REGIONS OF OPERATION

REGION	CONDITION	DRAIN CURRENT
CUTOFF	$v_{GS} \leq V_t$	$i_D = 0$
TRIODE	$v_{GS} > V_t, v_{DS} < v_{GS} - V_t$	$i_D = k'_n \frac{W}{L} \left[(v_{GS} - V_t)v_{DS} - \frac{1}{2}v_{DS}^2 \right]$
SATURATION	$v_{GS} > V_t, v_{DS} \geq v_{GS} - V_t$	$i_D = \frac{1}{2}k'_n \frac{W}{L} (v_{GS} - V_t)^2$

- In a DC problem: **assume** a region, **solve**, then **check** the assumption.
- Current saturates because the channel is pinched off at the drain end.